

eBay E-Commerce Sales Dashboard

Business Problem Statement & Project Deliverables

Dataset: eBay Sales
2024

Records: 13,000
Orders

Tool: Microsoft
Excel

Cities: 12 US Cities

Categories: 5
Product Lines

Business Problem Statement

eBay, a global e-commerce marketplace, has accumulated a full year of transactional sales data across 13,000 orders spanning 12 major US cities and 5 product categories — Electronics, Clothing, Home, Office, and Sports. While the raw data exists, it remains unstructured, inconsistent, and unanalyzed, making it impossible for management to draw reliable conclusions or make data-driven decisions.

The management team has observed fluctuating revenue patterns across different months, cities, and product categories. They are unable to identify which customer segments generate the most value, which product lines are underperforming, or what geographic opportunities exist for targeted marketing and expansion.

"How can eBay leverage its 2024 consumer sales data to identify high-value customer segments, optimize product category performance, and enable data-driven decisions across 12 US markets?"

Key Business Challenges

1

Raw Data Quality

13,000 rows of messy, unstructured transactional data with inconsistent formats, missing values, and unclassified order statuses requiring thorough cleaning before any analysis.

2

Revenue Visibility

No consolidated view of total revenue (\$5.74M), category-wise breakdown, or month-over-month trends, leaving management unable to assess business performance at a glance.

3

Customer Segmentation Gap

No system exists to classify customers into value tiers (Bronze, Silver, Gold) despite wide revenue variance (\$242K–\$340K per customer), making VIP retention impossible.

4

Geographic Blind Spot

Sales data spans 12 cities (NYC, LA, CHI, SF, BOS, MIA, ATL, DAL, DEN, PHX, PDX, SEA) with no comparative analysis of city-level AOV or revenue contribution.

5

Product Category Insight

Electronics dominates revenue (\$3.37M, 59%) but no cross-analysis exists between categories and months to identify seasonal demand patterns or slow-moving inventory.

Project KPI Snapshot (From Your Dashboard)

TOTAL REVENUE	TOTAL ORDERS	AVG ORDER VALUE	MoM GROWTH	TOP CATEGORY
\$5,744,894	13,000	\$457.97	+44.2%	Electronics
Full Year 2024	Validated Orders	Per Transaction	\$93K → \$134K	59% of Revenue

Project Deliverables

1 Data Cleaning & Preparation (Excel)

- ✓ Cleaned 13,000 rows of raw eBay transactional data
- ✓ Split combined fields, standardized dates, removed duplicates
- ✓ Created Revenue column (Qty x Price), Month column, Status validation
- ✓ Built CustomerTotalRevenue and CustomerTier (Bronze/Silver/Gold) columns using SUMIF + INDEX-MATCH
- ✓ Output: eBay-Cleaned CSV sheet — production-ready dataset

2 PivotTable Analysis Engine (Excel)

- ✓ PivotTable 1 — Revenue by Category: Electronics (\$3.37M), Clothing, Home, Office, Sports
- ✓ PivotTable 2 — Monthly Trends: 12-month revenue and order count tracking (Jan–Dec 2024)
- ✓ PivotTable 3 — Top 10 Customers by Revenue (Charlie: \$340K, Ivy: \$339K, Tina: \$322K...)
- ✓ PivotTable 4 — Product Matrix: Category x Month cross-tab heatmap
- ✓ All PivotTables connected via Slicers (Category, Status, City)

3 Advanced Formula KPIs (Excel)

- ✓ Total Revenue: \$5,744,894 using SUMIFS on validated orders
- ✓ Orders Count: 13,000 using SUMPRODUCT
- ✓ Average Order Value (AOV): \$457.97
- ✓ MoM Growth: +44.2% (Previous: \$93,448 → Current: \$134,773)
- ✓ Customer Tier Classification using INDEX-MATCH with percentile-based thresholds

4 Interactive Dashboard (Excel)

- ✓ Single-page executive dashboard with blue theme (#2E5A8D / #F4F7FA)
- ✓ 6 dynamic KPI cards with conditional formatting (green/red indicators)
- ✓ Revenue Trend chart (Line + Bar combo across 12 months)
- ✓ Category Donut chart showing revenue share by product line
- ✓ Top 10 Customers horizontal bar chart
- ✓ Category x Month heatmap with color-scale conditional formatting
- ✓ Interactive slicers connected to all PivotTables simultaneously